

Executive Business Report

Superstore Sales Forecasting and Demand Analytics

Executive Summary

This report presents a comprehensive analysis of four years of Superstore sales data to support inventory planning, demand forecasting, and strategic decision-making. Historical sales patterns were examined to identify trends, seasonal demand fluctuations, unusual sales events, and product demand behavior. Three forecasting models were evaluated, with the XGBoost model achieving the highest forecasting accuracy. The analysis indicates continued sales growth over the next three months, particularly within the Technology category and the West region. In addition, anomaly detection revealed several significant sales spikes and declines that likely correspond to seasonal promotions and operational fluctuations. Product demand segmentation further identified distinct inventory groups, enabling targeted stocking strategies that can improve product availability while reducing excess inventory costs.

Key Findings

Exploratory Data Analysis

The analysis revealed that the Technology category generated the highest overall revenue, making it the most important contributor to total sales. Monthly sales exhibited clear seasonality, with demand consistently increasing toward the end of each year, particularly during November and December. The average shipping time remained relatively stable across all regions, indicating a consistent fulfillment process. Overall sales demonstrated a positive long-term growth trend throughout the four-year period despite periodic fluctuations.

Time Series Analysis

Time series decomposition showed a clear upward sales trend combined with recurring seasonal patterns. Statistical testing confirmed that the sales series was sufficiently stable for forecasting purposes. This indicates that historical sales behavior provides a reliable foundation for future demand prediction.

Forecasting Model Evaluation

Three forecasting approaches were developed and compared: SARIMA, Facebook Prophet, and XGBoost.

Model	MAE	RMSE	MAPE
SARIMA	20,580.70	22,190.91	21.94%
Facebook Prophet	20,250.79	22,318.41	21.86%
XGBoost	14,763.81	18,337.41	14.48%

Based on these evaluation metrics, XGBoost achieved the lowest prediction errors across all measures and was therefore selected as the preferred forecasting model.

Three-Month Sales Forecast

Using the XGBoost forecasting model, sales are expected to remain stable with continued positive growth over the next three months.

Forecast Month	Predicted Sales
Month 1	86,530
Month 2	87,955
Month 3	88,256

Considering the model's historical forecasting error (RMSE $\approx$ 18,337), actual sales may vary around these predictions. However, the forecast indicates a positive business outlook with no significant decline in demand expected during the forecast period.

Category and Region Forecast Insights

Additional forecasting was performed for major product categories and regions using the best-performing XGBoost model.

Projected growth over the forecast horizon:

Segment	Projected Growth
West Region	10,181.94
Technology	9,201.06
Furniture	6,954.17
East Region	3,464.33
Office Supplies	3,215.77

Among all analyzed segments, the West region is expected to experience the strongest growth, followed closely by the Technology category. These segments represent the most significant opportunities for future revenue expansion and should receive priority in inventory planning.

Sales Anomalies

Two complementary anomaly detection techniques were used to identify unusual sales activity.

The most significant anomalies included:

- **22 March 2015:** Weekly sales reached 37,703.67, representing the largest sales spike detected in the dataset. This event may have been driven by a promotional campaign, bulk customer purchases, or seasonal demand.
- **November–December 2018:** Multiple unusually high sales weeks were identified, likely corresponding to Black Friday, Cyber Monday, and year-end holiday shopping activity.
- **Several low-sales weeks during early 2015 and early 2016:** These periods may reflect post-holiday demand slowdowns, inventory shortages, or operational disruptions.

Isolation Forest identified 11 anomalous weeks, while the rolling Z-score method detected 6 anomalous weeks. The differences between the two methods highlight that unusual business events can be identified from multiple analytical perspectives.

Product Demand Segmentation

Product sub-categories were grouped into four demand segments using clustering analysis.

Premium High-Value Products

Sub-categories: Copiers, Machines

These products generate high revenue per transaction and require careful inventory management to balance product availability against holding costs.

High Volume, Stable Demand

Sub-categories: Accessories, Binders, Chairs, Phones, Storage, Tables

These products generate consistent demand and contribute significantly to overall revenue. Inventory availability should be prioritized to avoid stock-outs.

Low Volume, Stable Demand

Sub-categories: Appliances, Bookcases, Envelopes, Furnishings, Fasteners, Art, Paper, Labels

These products exhibit predictable purchasing behavior and can be managed through routine replenishment schedules.

Growing Demand

Sub-category: Supplies

This segment demonstrated exceptionally strong year-over-year growth and should be monitored closely for future expansion opportunities.

Business Recommendations

1. Increase Inventory Investment in Technology Products and the West Region

The Technology category and West region demonstrated the strongest projected growth. Prioritizing inventory availability in these segments will help capture future demand and maximize revenue opportunities.

2. Strengthen Inventory Planning Before Year-End Peak Seasons

Historical sales data consistently showed strong demand during November and December. Additional inventory, staffing, and logistics capacity should be allocated ahead of these periods to avoid stock shortages and fulfillment delays.

3. Adopt XGBoost-Based Forecasting for Operational Planning

Among all evaluated forecasting approaches, XGBoost achieved the lowest forecasting error with a MAPE of 14.48%. Using this model for short-term planning can improve purchasing decisions, inventory allocation, and demand management.

Business Risk and Limitation

The forecasting system relies primarily on historical sales behavior. Unexpected events such as economic changes, supply chain disruptions, competitor actions, or major marketing campaigns may significantly influence future demand and reduce forecast accuracy. To maintain reliable predictions, the forecasting model should be retrained regularly using the latest available sales data.

Conclusion

This project demonstrates how data analytics and machine learning can support demand forecasting, inventory optimization, and strategic supply chain management. By combining forecasting, anomaly detection, and product demand segmentation, the proposed solution provides actionable insights that can help improve product availability, reduce operational costs, and support data-driven business decisions.